

Assignment No. 6

Title:

Perform Data Clustering Using Any Clustering Algorithm

Objectives:

- To understand the concept of clustering in machine learning
- To implement a clustering algorithm on a dataset
- To analyze grouping of similar data points

Problem Statement:

Perform data clustering using any clustering algorithm and group similar data points based on their features.

Software Requirements:

- Python (Anaconda / Jupyter Notebook) or Google Colab
- Libraries: NumPy, Matplotlib, Scikit-learn

Hardware Requirements:

- Processor: Minimum 1 GHz
- Hard Drive: Minimum 32 GB
- Memory (RAM): Minimum 2 GB

Theory

Introduction

Clustering is an unsupervised machine learning technique used to group similar data points into clusters based on their characteristics. Unlike classification, clustering does not use labeled data. It helps in identifying hidden patterns or structures within data.

Clustering Concept

In clustering:

- Data points in the same cluster are highly similar
- Data points in different clusters are dissimilar

Clustering is widely used in data analysis, pattern recognition, and business intelligence.

Clustering vs Classification

- **Clustering:** Unsupervised learning, no predefined labels
- **Classification:** Supervised learning, uses labeled data

K-Means Clustering Algorithm

K-Means is a popular clustering algorithm that partitions the dataset into **K clusters**. Each cluster is represented by its centroid (mean value of points in that cluster).

Steps of K-Means Algorithm

1. Choose the number of clusters (K)
2. Initialize centroids randomly
3. Assign each data point to the nearest centroid
4. Update centroids by calculating mean
5. Repeat until convergence

Applications of Clustering

- Customer segmentation

- Market analysis
- Image segmentation
- Recommendation systems
- Anomaly detection

Implementation / Procedure:

- Import required libraries
- Load or create dataset
- Apply K-Means clustering algorithm
- Fit the model to the data
- Predict cluster labels
- Visualize clusters using graphs

Python Implementation

```
# Import libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Dataset
X = np.array([[1,2], [1,4], [1,0],
              [10,2], [10,4], [10,0]])

# Apply K-Means
kmeans = KMeans(n_clusters=2)
kmeans.fit(X)

# Results
labels = kmeans.labels_
centroids = kmeans.cluster_centers_

# Visualization
plt.scatter(X[:,0], X[:,1], c=labels)
plt.scatter(centroids[:,0], centroids[:,1], marker='X')
plt.title("K-Means Clustering")
plt.show()
```

Configuration Steps:

- Install required libraries (NumPy, Matplotlib, Scikit-learn)
- Set up Python environment (Jupyter Notebook / Colab)

- Load dataset properly
- Configure number of clusters (K value)
- Execute code and verify output

Conclusion:

In this assignment, we successfully implemented a clustering algorithm using K-Means. The algorithm grouped similar data points into clusters based on their features. Clustering is useful for discovering hidden patterns and plays an important role in data analysis and machine learning applications.